

Implementation and validation of nonlinear spring–damper restraints in OpenFOAM’s six-DOF rigid-body motion framework

Related article

These directories and files are related to the following article:

Poorya Poozesh, Antonio J. Álvarez, Arturo N. Fontán, and Félix Nieto, *Implementation and validation of nonlinear spring–damper restraints in OpenFOAM’s six-DOF rigid-body motion framework*, OpenFOAM Journal, 2026.

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Overview

This repository contains two custom six-degree-of-freedom (six-DOF) restraint libraries for OpenFOAM:

- nonLinearSpring: nonlinear translational spring-damper restraint.
- nonLinearAxialAngularSpring: nonlinear axial/angular spring-damper restraint.

Both classes derive from sixDoFRigidBodyMotionRestraint and are available through the OpenFOAM runtime-selection table.

Supported spring models are:

- linear: $F = A \cdot x$, where x is displacement.
- Duffing: $F = A \cdot x + B \cdot x^3$.
- stiffnessList: a tabulated displacement-force or rotation-moment relation.

Supported damper models are:

- linear: $F = c1 \cdot v$, where v is velocity.
- Rayleigh: $F = c1 \cdot v + c2 \cdot v^3$.
- Mann: $F = c1 \cdot v + c2 \cdot x^2 \cdot v$.
- dampingList: a tabulated displacement-force or velocity-force relation and its rotational equivalent.

Tabulated stiffness and damping data can use linear or cubic-spline interpolation. The repository also includes two libraries that provide linear extrapolation outside the tabulated range:

- libinterExtrapolationTable: linear interpolation with linear extrapolation.
- libinterExtrapolateSplineXY: cubic-spline interpolation with linear extrapolation.

Compatibility and requirements

- OpenFOAM ESI must be loaded before compiling or running a case. The supplied code has been tested with OpenFOAM v2212 and v2406.
- The tutorial cases are configured for eight subdomains in system/decomposeParDict.

- The plotting scripts require Python 3, NumPy, and Matplotlib. The os, sys, and copy modules used by the scripts are part of the Python standard library.
- Run the plotting scripts in a normal Python environment after the OpenFOAM run. A Python installation loaded inside the OpenFOAM environment can conflict with OpenFOAM's MPI libraries.

Repository layout

- Libraries/: restraint and interpolation/extrapolation source code, plus Allwmake and Allclean.
- Tutorials/Translational/freeDecay/: cases 01-04.
- Tutorials/Translational/wind/: cases 05-07.
- Tutorials/Rotational/freeDecay/: cases 08-10.
- Tutorials/Rotational/wind/: case 11.
- Tutorials/FieldsForMapping/: steady fields mapped by the wind-excited wing cases to reduce the initial flow transient.
- FSI_applicationExample/: case 12, the 3:2 rectangular-prism application.

Compile the libraries

From a shell in which OpenFOAM ESI is loaded:

```
cd Libraries
./Allwmake
```

The compiled libraries are written to \$FOAM_USER_LIBBIN. To remove the installed libraries and generated lnInclude directories, keep OpenFOAM loaded and run:

```
./Allclean
```

Before creating a distribution archive, also remove the platform-specific Make/<platform> directories and their object/dependency files by running wclean libso in every library directory:

```
for library in interExtrapolateSplineXY interExtrapolationTable nonLinearSpring nonLinearAxialAngularSpring
do
(cd "$library" && wclean libso)
done
```

Run and post-process a case

Each case contains an Allrun script that generates the mesh, prepares the initial fields, decomposes the domain, and runs the configured solver in parallel. Wind-excited wing cases also map the initial solution from Tutorials/FieldsForMapping.

From the selected case directory:

```
./Allrun
```

After the simulation finishes, run the case-specific plotting script from the same directory, preferably in a separate shell with a standard Python environment:

```
python3 plotGeneration.py
```

The script reads the rigid-body state from `postProcessing/sixDoFRigidBodyState/0/sixDoFRigidBodyState.dat`. Wind-excited cases also read `postProcessing/forceCoeffsWing/0/coefficient.dat`. If OpenFOAM writes to a start-time directory other than 0, update those paths in `plotGeneration.py`.

Every supplied plotting script writes:

- image.pdf
- image.png

For free-decay cases, these files contain the displacement or rotation time history. For wind-excited cases, they contain the motion and the relevant force coefficient (C_l for heave and C_m for pitch). The case-12 plot contains y/H and C_l histories.

Case-to-manuscript mapping

The mapping below refers to the manuscript titled *Implementation and validation of nonlinear spring–damper restraints in OpenFOAM’s six-DOF rigid-body motion framework*. A case reproduces the OpenFOAM-library contribution to the cited result. Comparative preCICE, standard-linearSpring, analytical, and central-finite-difference datasets are not included in this archive unless stated otherwise.

Free-decay and verification cases

Case	Manuscript result	Active DOF	Spring and damper setup	Interpolation	Default timeStep	Main supplied output
01_wing_linearSpring_linearDamping	Section 4.1; Figure 4; Tables 2, 3, and 6	Heave, global y translation (reported as x in the manuscript)	Linear stiffness represented by stiffnessList, $k = 4000$ N/m; linear damping, $c1 = 6$ N s/m	Linear; EXTRAPOLATE	1.0e-5 s	Translational free-decay history in image.pdf and image.png
02_wing_listSpring_linearDamping	Companion free-decay case for the tabulated nonlinear law used in Section 5.2 and Figure 13; no dedicated response figure or table	Heave, global y translation	Nonlinear stiffnessList; linear damping, $c1 = 6$ N s/m	Linear; EXTRAPOLATE	1.0e-6 s	Translational free-decay history; the stiffness list itself defines the Figure 13 law
03_wing_DuffingSpring_noDamping	Section 4.2; Figure 6; Tables 7, 8, and 11	Heave, global y translation	Duffing stiffness, $A = 4000$ N/m, $B = 200000$ N/m ³ ; zero damping ($c1 = c2 = 0$)	Not applicable	1.0e-6 s	Translational undamped history in image.pdf and image.png
04_wing_DuffingSpring_MannDamping	Section 4.3; Figure 8; Tables 12 and 14	Heave, global y translation	Duffing stiffness, $A = 4000$ N/m, $B = 200000$ N/m ³ ; Mann damping, $c1 = 6$, $c2 = 5000$	Not applicable	1.0e-6 s	Translational nonlinear damped history; the central-difference reference is not generated by the supplied script
08_wing_linearSpring_linearDamping	Section 4.1; Figure 5; Tables 4, 5, and 6	Pitch, rotation about global z	Linear stiffness and damping represented by lists, $k = 8000$ N m/rad, $c = 20$ N m s/rad	Linear; EXTRAPOLATE	1.0e-5 s	Rotational free-decay history in image.pdf and image.png
09_wing_Duffing_noDamping	Section 4.2; Figure 7; Tables 9, 10, and 11	Pitch, rotation about global z	Duffing stiffness, $A = 8000$ N m/rad, $B = 200000$ N m/rad ³ ; zero damping ($c1 = c2 = 0$)	Not applicable	1.0e-6 s	Rotational undamped history in image.pdf and image.png
10_wing_listSpring_RayleighDamping	Section 4.3; Figures 9 and 10; Tables 13 and 14	Pitch, rotation about global z	Nonlinear stiffnessList; Rayleigh damping, $c1 = 20$, $c2 = 50$	Linear; EXTRAPOLATE	1.0e-6 s	Rotational nonlinear damped history; the central-difference reference is not generated by the supplied script

Wind-excited and application cases

Case	Manuscript result	Active DOF	Spring and damper setup	Interpolation	Default Δt	Main supplied output
05_wing_linearSpring_linearDamping	Section 5.1; Figures 11 and 12	Heave, global y translation (reported as x in the manuscript)	Linear stiffness and velocity damping represented by lists, $k = 4000 \text{ N/m}$, $c = 2 \text{ N s/m}$	Linear; EXTRAPOLATE	$1.0\text{e-}5 \text{ s}$	Non-dimensional heave and lift-coefficient histories; preCICE and standard-linearSpring series are not included
06_wing_listSpring_linearDamping	Section 5.2; Figures 13 and 14	Heave, global y translation	Nonlinear stiffnessList; linear damping, $c1 = 2 \text{ N s/m}$	Linear; EXTRAPOLATE	$1.0\text{e-}6 \text{ s}$	Non-dimensional heave and lift-coefficient histories; the preCICE series is not included
07_wing_DuffingSpring_MannDamping	Section 5.3; Figures 15 and 16; Table 15	Heave, global y translation	Duffing stiffness, $A = 4000 \text{ N/m}$, $B = 200000 \text{ N/m}^3$; Mann damping, $c1 = 6$, $c2 = 5000$	Not applicable	$1.0\text{e-}5 \text{ s}$	Non-dimensional heave and lift-coefficient histories
11_wing_listSpring_RayleighDamping	Section 5.4; Figure 17	Pitch, rotation about global z	Nonlinear stiffnessList; Rayleigh damping, $c1 = 20$, $c2 = 50$	Linear; EXTRAPOLATE	$1.0\text{e-}6 \text{ s}$	Pitch and moment-coefficient histories
12_prism3_2_listSpring_linearDamping	Section 6; setup in Figure 2 and Tables 16-19; results in Figures 18-23 and Tables 20-21	Heave, global y translation	Nonlinear stiffnessList; linear damping, $c1 = 4.594375 \text{ N s/m}$	Linear; EXTRAPOLATE	$8.0\text{e-}4 \text{ s}$	y/H and lift-coefficient histories; mesh, spectra, phase averages, statistics, Q criterion, and preCICE comparison require additional post-processing/data

Time-step sensitivity runs

The archive provides one default deltaT per case. The manuscript figures and tables listed below combine multiple simulations. To reproduce them, copy the relevant case to a separate working directory for each time step, change deltaT in system/controlDict, and run ./Allclean followed by ./Allrun. Preserve each run's postProcessing directory and log files before starting another run.

Manuscript result	Base cases	Required time steps
Figures 4 and 5; Tables 3, 5, and 6	01 and 08	7.2e-5, 1.0e-5, and 1.0e-6 s
Figures 6 and 7; Tables 8, 10, and 11	03 and 09	7.2e-5, 1.0e-5, and 1.0e-6 s
Figure 11	05	7.2e-5, 1.0e-5, and 1.0e-6 s
Figure 14	06	1.0e-5 and 1.0e-6 s for the OpenFOAM nonLinearSpring curves
Figure 15	07	7.2e-5, 1.0e-5, and 1.0e-6 s

Cases 04 and 10 use 1.0e-6 s for the Section 4.3 numerical-verification comparison. The application-case values shown in the mapping table are the defaults supplied in the archive.

Final validation and archive cleanup

Before distributing the repository, build the libraries and run the required tutorial checks in a clean OpenFOAM v2406 environment. Use the same OpenFOAM environment for compilation and execution, and confirm that the custom restraint libraries are found in \$FOAM_USER_LIBBIN and loaded by the tutorial controlDict files.

After verification, run ./Allclean in every tutorial or application case that was executed. This removes generated processor directories, reconstructed initial fields, meshes, post-processing results, logs, and plots. Preserve the original 0.org directories. The supplied Tutorials/FieldsForMapping/0 and Tutorials/FieldsForMapping/constant/polyMesh directories are intentional input data used by the wind-excited cases and must also be preserved.

Run the library cleanup described above, and then check the release tree for remaining compilation or execution products:

```
find Libraries -type d \( -name 'linux*' -o -name 'lnInclude' \) -print
find Libraries -type f \( -name '*.o' -o -name '*.dep' -o -name '*.so' \) -print
find Tutorials FSI_applicationExample -type d \( -name 'processor*' -o -name 'postProcessing' \) -print
find Tutorials FSI_applicationExample -type f \( -name 'log*' -o -name 'image.pdf' -o -name 'image.png' \) -print
```

These commands should produce no output. Remove any reported generated files or directories before packaging the archive. Also remove reconstructed numerical time directories and generated constant/polyMesh directories from ordinary case folders, while retaining the FieldsForMapping input data identified above.

FSI application resource note

FSI_applicationExample/12_prism3_2_listSpring_linearDamping is substantially more expensive than the wing tutorials. Its Allrun script uses the number of subdomains set in system/decomposeParDict (eight in the supplied archive), but a publication-quality production run may require more cores and several days. Adjust the decomposition to the available hardware and record the OpenFOAM version, core count, time step, end time, and any dictionary changes with the reproduced results.

Disclaimer

The nonlinear spring-damper libraries are provided as is, without warranty. The developers are not liable for errors, failures, or damages arising from their use. Users assume full responsibility for applying the code in their own work.